

Donggyu Kim (김동규)

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Current Positions

09/2026– Assistant Professor, Department of Mathematics, **Hanyang University**, Seoul, South Korea.

Research Areas

Algebraic Combinatorics and Matroid Theory.
In particular, algebraic properties of graphs; matroids with coefficients; Coxeter matroids.

Previous Positions

03/2025–07/2026 Visiting Assistant Professor (08/2025–07/2026) and Postdoctoral Fellow (03/2025–07/2025)
School of Mathematics, **Georgia Institute of Technology**, Atlanta, GA, USA.
Postdoc mentor: Prof. **Matthew Baker**.

07/2019–02/2025 Student researcher (학생 연구원), Discrete Mathematics Group, **IBS** (Institute for Basic Science), Daejeon, Korea.

Education

03/2019–02/2025 **KAIST**, Daejeon, Korea.
Ph.D. in Mathematical Sciences, February 2025.
Thesis title: *Delta-matroids with coefficients and linear spaces equipped with a bilinear form*.
Thesis advisor: Prof. **Sang-il Oum**.

03/2015–02/2019 **KAIST**, Daejeon, Korea.
B.S. in Mathematical Sciences with *Cum Laude*.

03/2012–02/2015 Korea Science Academy of KAIST (한국과학영재학교), Busan, Korea.

Awards

Nov. 26, 2025 **the 15th S-OIL Excellent Dissertation Award** (제15회 เอส-오일 우수학위 논문상), S-OIL 과학문화재단.

July 2025–June 2027 AMS-Simons Travel Grant, American Mathematical Society (funded by the Simons Foundation).

May 28, 2022 Excellent Poster Award, 2022 KSIAM Spring Conference.

Journal Papers

1. Matthew Baker, Changxin Ding, and **Donggyu Kim**. “The Jacobian of a regular orthogonal matroid and torsor structures on spanning quasi-trees of ribbon graphs”.

- Adv. Math.*, September 2026. DOI: [10.1016/j.aim.2026.111099](https://doi.org/10.1016/j.aim.2026.111099), arXiv: [2501.08796](https://arxiv.org/abs/2501.08796).
2. Tong Jin and **Donggyu Kim**. “Orthogonal matroids over tracts”. *Forum Math. Sigma*, August 2025. DOI: [10.1017/fms.2025.10085](https://doi.org/10.1017/fms.2025.10085), arXiv: [2303.05353](https://arxiv.org/abs/2303.05353).
 3. **Donggyu Kim**. “Baker–Bowler theory for Lagrangian Grassmannians”. *Int. Math. Res. Not. IMRN*, April 2025. DOI: [10.1093/imrn/rnaf094](https://doi.org/10.1093/imrn/rnaf094), arXiv: [2403.02356](https://arxiv.org/abs/2403.02356).
 4. **Donggyu Kim** and Sang-il Oum. “Note on Hamiltonicity of basis graphs of even delta-matroids”. *J. Graph Theory*, March 2025. DOI: [10.1002/jgt.23237](https://doi.org/10.1002/jgt.23237), arXiv: [2308.05772](https://arxiv.org/abs/2308.05772).
 5. Junggho Ahn, Debsoumya Chakraborti, Kevin Hendrey, **Donggyu Kim**, and Sang-il Oum. “Twin-width of random graphs”. *Random Structures Algorithms*, June 2024. DOI: [10.1002/rsa.21247](https://doi.org/10.1002/rsa.21247), arXiv: [2212.07880](https://arxiv.org/abs/2212.07880).
 6. **Donggyu Kim** and Sang-il Oum. “Vertex-minors of graphs: A survey”. *Discrete Appl. Math.*, July 2024. DOI: [10.1016/j.dam.2024.03.011](https://doi.org/10.1016/j.dam.2024.03.011).
 7. **Donggyu Kim** and Sang-il Oum. “Prime vertex-minors of a prime graph”. *European J. Combin.*, **118**:103871, May 2024. DOI: [10.1016/j.ejc.2023.103871](https://doi.org/10.1016/j.ejc.2023.103871), arXiv: [2202.07877](https://arxiv.org/abs/2202.07877).
 8. **Donggyu Kim**, Duksang Lee, and Sang-il Oum. “T-graphic delta-matroids and their applications”. *Combinatorica*, **43**(5):963–983, October 2023. DOI: [10.1007/s00493-023-00043-6](https://doi.org/10.1007/s00493-023-00043-6), arXiv: [2104.11383](https://arxiv.org/abs/2104.11383).
 9. **Donggyu Kim** and Suil O. “Eigenvalues and parity factors in graphs with given minimum degree”. *Discrete Math.*, **346**(4):113290, April 2023. DOI: [10.1016/j.disc.2022.113290](https://doi.org/10.1016/j.disc.2022.113290), arXiv: [2111.12966](https://arxiv.org/abs/2111.12966).
 10. Junggho Ahn, Kevin Hendrey, **Donggyu Kim** and Sang-il Oum. “Bounds for the twin-width of graphs”. *SIAM J. Discrete Math.*, **36**(3):2352–2366, 2022. DOI: [10.1137/21M1452834](https://doi.org/10.1137/21M1452834), arXiv: <https://arxiv.org/abs/2110.03957>.

Submitted Papers / Preprints

11. Seonghyuk Im and **Donggyu Kim**. “On the enumeration of polymatroids”. arXiv: [2608.18225](https://arxiv.org/abs/2608.18225), 2026.
12. Changxin Ding and **Donggyu Kim**. “A polyhedral approach to homotopy theorems in matroid theory”. arXiv: [2607.22993](https://arxiv.org/abs/2607.22993), 2026.
13. Changxin Ding and **Donggyu Kim**. “Pseudo-orientable ribbon graphs: Matrix–Quasi-tree Theorem and log-concavity”. arXiv: [2603.05702](https://arxiv.org/abs/2603.05702), 2026.
14. **Donggyu Kim**, Rose McCarty, and Caleb McFarland. “Graphs whose Eulerian trails have unique labels”. arXiv: [2603.02501](https://arxiv.org/abs/2603.02501), 2026.
15. Changxin Ding and **Donggyu Kim**. “Grassmann–Plücker functions for orthogonal matroids”. arXiv: [2601.19830](https://arxiv.org/abs/2601.19830), 2026.
16. Matthew Baker, June Huh, **Donggyu Kim**, Mario Kummer, Oliver Lorscheid. “Representation theory for polymatroids”. arXiv: [2507.14718](https://arxiv.org/abs/2507.14718), 2025.
17. Rutger Campbell, **Donggyu Kim**, and Jorn van der Pol. “Turán’s theorem for Dowling geometries”. arXiv: [2508.20843](https://arxiv.org/abs/2508.20843), 2025.

18. **Donggyu Kim** and Hyunwoo Lee. “Reconstructing hypergraph matching polynomials”. arXiv: [2501.19081](#), 2025.

Refereed Conference Papers

19. **Donggyu Kim**, Duksang Lee, and Sang-il Oum. “T-graphic delta-matroids and their applications”. In: *Proceedings of the 32nd International Symposium on Algorithms and Computation (ISAAC2021, December 6-8, 2021, Fukuoka, Japan)*. Ed. by Hee-Kap Ahn and Kunihiro Sadakane. Vol. 212. Accepted. Dagstuhl, Germany: Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2021, 70:1-70:13. DOI: [10.4230/LIPIcs.ISAAC.2021.70](#).

Research Talks

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| Oct. 13, 2026 | TBA, <i>Colloquium</i> , National Yang Ming Chiao Tung University, Hsinchu, Taiwan. |
| Sep. 5, 2026 | TBA, <i>Colloquium</i> , Hanyang University, Seoul, South Korea. |
| June 30, 2026 | Unimodular representations of ribbon graphs, <i>SiMEGG: Structure in Matroids, Embedded Graphs and Graphs</i> , ICMS, Bayes Centre, Edinburgh, UK. |
| June 18, 2026 | Theory of matroids with coefficients and its generalizations, <i>Special Seminar</i> , Sungkyunkwan University, Suwon, South Korea. |
| May 17, 2026 | Grassmann–Plücker functions for orthogonal matroids, <i>34th Cumberland Conference</i> , Auburn University, Auburn, AL, U.S. |
| Mar. 29, 2026 | Pseudo-orientable ribbon graphs: Matrix–Quasi-tree Theorem and log-concavity, <i>Special session on Algebraic Combinatorics at the AMS Sectional Meeting</i> , Georgia Southern University in Savannah, GA, U.S. |
| Feb. 25, 2026 | Pseudo-orientable ribbon graphs: Matrix–Quasi-tree Theorem and log-concavity, <i>Combinatorics Seminar</i> , Auburn University, Auburn, AL, U.S. |
| Jan. 8, 2026 | Representation theory for polymatroids, <i>Combinatorics Seminar at HY</i> , Hanyang University, Seoul, Korea. |
| Dec. 26, 2025 | Grassmann–Plücker functions for orthogonal matroids, <i>Discrete Math Seminar</i> , IBS Discrete Mathematics Group, Daejeon, South Korea. |
| Dec. 22, 2025 | Representation theory for polymatroids, 35th KIAS Combinatorics Workshop , Busan, South Korea. |
| Sep. 12, 2025 | Turán’s theorem for Dowling geometries, <i>Combinatorics Seminar</i> , Georgia Tech, Atlanta, Georgia, U.S. |
| May 21, 2025 | Group structures on spanning quasi-trees of ribbon graphs and regular orthogonal matroids, <i>CanadAM 2025</i> , University of Ottawa, Ottawa, Canada. |
| Feb. 28, 2025 | Torsor structures on spanning quasi-trees of ribbon graphs, <i>Discrete Analysis Seminar</i> , Yonsei University, Seoul, South Korea. |
| Oct. 24, 2024 | Baker-Bowler theory for Lagrangian Grassmannians, Geometry of Matroids Workshop , IAS, Princeton, New Jersey, U.S. |

- Aug. 30, 2024 Two ways to generalize matroids with coefficients, *2024 Combinatorics Workshop*, Chung Buk National University, Cheongju, South Korea.
- Aug. 20, 2024 Baker-Bowler theory for Lagrangian Grassmannians, *2024 Workshop on (Mostly) Matroids*, IBS, Daejeon, South Korea.
- Jul. 8, 2024 Baker-Bowler theory for Lagrangian Grassmannians, *SIAM Conference on Discrete Mathematics (DM24)*, Spokane, Washington, U.S.
- Apr. 19, 2024 Baker-Bowler theory for Lagrangian Grassmannians, *2024 KMS Spring Meeting*, Daejeon, South Korea.
- Apr. 15, 2024 Baker-Bowler theory for Lagrangian Grassmannians, *Joint seminar Atlanta-Groningen*, Online.
- Mar 18, 2024 Twin-width of random graphs, *Combinatorics Seminar*, Yeungnam University, Gyeongsan, Gyeongbuk, South Korea.
- Mar 9, 2024 Orthogonal matroids over tracts, **30th KIAS Combinatorics Workshop**, Seoul, South Korea.
- Oct. 28, 2023 Hamiltonicity of basis graphs of even delta-matroids, *2023 KMS Annual Meeting*, Seoul, South Korea.
- Sep. 19, 2023 Orthogonal matroids over tracts, *Discrete Math Seminar*, IBS Discrete Mathematics Group, Daejeon, South Korea.
- Jun. 7, 2023 Prime vertex-minors of prime graphs, *CanADAM 2023*, Winnipeg, Canada.
- Apr. 29, 2023 Orthogonal matroids over tracts, *2023 KMS Spring Meeting*, Daejeon, South Korea.
- April, 2023 (Poster) Orthogonal Matroids over Tracts, (Internal) IBS ECOPRO and DIMAG poster day.
- Dec. 4, 2022 Matroid theory arising from projective geometry, *Workshop for Young Mathematicians in Korea 2022*, KAIST, Daejeon, South Korea.
- Jun. 1, 2022 Stronger versions of Tutte's wheel theorem for vertex-/pivot-minors, *Student Symposium in Combinatorics 2022*, Online, Australia.
- May 28, 2022 (Poster) Γ -graphic delta-matroids and their applications, *2022 KSIAM Spring Conference*, IBS, Daejeon, South Korea.
- Apr. 28, 2022 A stronger version of Tutte's wheel theorem for vertex-minors, *2022 KMS Spring Meeting*, Online, South Korea.
- Feb. 21, 2022 A stronger version of Tutte's wheel theorem for vertex-minors, *Discrete Math Seminar*, IBS Discrete Mathematics Group, Daejeon, South Korea.
- Nov. 25, 2021 Eigenvalues and parity factors in graphs, *KAIST Math Graduate student Seminar*, Online, KAIST, South Korea.
- Nov. 14, 2021 Eigenvalues and parity factors in graphs, *2021 China-Korea Young Scholars' Symposium on Discrete Mathematics, Statistics and Optimization*, Online, South Korea and China.
- Oct. 26, 2021 Γ -graphic delta-matroids and their applications, *Discrete Math Seminar*, IBS Discrete Mathematics Group, Daejeon, South Korea.
- Oct. 21, 2021 Γ -graphic delta-matroids and their applications, *2021 KMS Annual*

Meeting, Online, South Korea.

Organizing Academic Events

- 2025 Fall–current Co-organizing the Algebra Seminar at Georgia Tech
 Jul. 29–Aug. 2, 2024 Co-organizing **2024 Korean Student Combinatorics Workshop**, Gongju, South Korea.

Teaching

- 2026 Spring Applied Combinatorics (Math3012 D/E), *Georgia Tech*.
 Directed Reading Program (mentored an undergraduate student through a semester-long reading project), *Georgia Tech*.
 2025 Fall Introduction to Linear Algebra (Math1553 B/C), *Georgia Tech*.
 2025 Summer Algebra, Summer Edge Program (intensive review of undergraduate-level abstract algebra for incoming graduate students), *Georgia Tech*.
 2023 Fall TA, Introduction to Graph Theory (MAS477), *KAIST*.
 2023 Summer TA, 2023 Third International Summer School for Advanced Undergraduate Students – Topological Methods in Combinatorics, *KAI-X Mathematics Global Workshop* (July 31st – August 3rd).
 2023 Spring TA, Linear Algebra (MAS212), *KAIST*.
 2022 Fall TA, Logic and Set Theory (MAS270), *KAIST*.
 2022 Spring TA, Discrete Mathematics (MAS275), *KAIST*.
 2021 Fall TA, Probability and Statistics (CC511), *KAIST*.
 TA, Introduction to Linear Algebra (MAS109), *KAIST*.
 2021 Spring TA, Differential Equations and Applications (MAS201), *KAIST*.
 TA, Discrete Mathematics (MAS275), *KAIST*.
 2020 Fall TA, Calculus I (MAS101), *KAIST*.
 TA, Introduction to Graph Theory (MAS477), *KAIST*.
 2020 Spring TA, Introduction to Linear Algebra (MAS109), *KAIST*. (**Excellent TA Award**)
 TA, Discrete Mathematics (MAS275), *KAIST*.
 2019 Fall TA, Introduction to Linear Algebra (MAS109), *KAIST*.
 TA, Linear Algebra (MAS212), *KAIST*.
 2019 Spring TA, Calculus I (MAS101), *KAIST*.

Review Activities

- Journals J. Combin. Theory Ser. A
 Bull. Iranian Math. Soc.
 European J. Combin.
 Algbr. Combin.
 Theoret. Comput. Sci.
 Discrete Appl. Math.
 J. Combin. Theory Ser. B
 Graphs Combin.
 Studia Sci. Math. Hungar.

Inform. and Comput.

Conferences 2026 FPSAC (2)
 2025 SODA
 2024 WG
 2023 WALCOM
 2023 MFCS

Math Posts

Apr. 27, 2026 Unimodular matrix representations for pseudo-orientable ribbon graphs - [A guest post in the Matroid Union](#)
Mar. 16, 2026 What are the cycles of ribbon graphs? - [A guest post in the Matroid Union](#)

Miscellaneous

Design KAIST 45th Anniversary Emblem Contest - [Prize winner \(Jan. 6th, 2016\)](#)
 KAIST Dept. of Mathematical Sciences Logo Design Contest - [2nd award \(Nov. 29th, 2021\)](#)

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